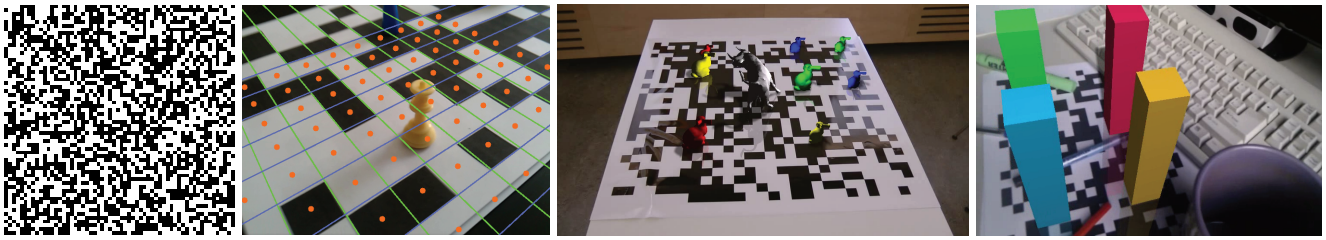
Uniform Marker Fields: Camera Localization By Orientable De Bruijn Tori

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ABSTRACT

In various applications, a wider area needs to be covered by fiduciary markers but a large marker cannot be used because only a fraction of the area is to be viewed by the camera. Such an area can be covered by a number of small markers with unique identifiers. However, with the camera freely moving in the scene and with occluders present, it is difficult to ensure that at least one of the individual markers is completely visible, unless the markers are small and numerous. In that case, the markers are not recognizable from larger distances.

In this paper we introduce the concept of *Marker Fields* which overcome this limitation. The Marker Field covers a large-scale planar (or non-planar) area and it is composed of mutually overlapping partial markers. We propose a particular arrangement of the Marker Field: a Uniform Checker-Board Marker Field, which is a black-and-white checkerboard whose square modules are defined by *aperiodic 4-orientable binary n^2-window arrays* (De Bruijn tori). We propose a genetic algorithm for construction of 4-orientable n^2-window arrays.

We used a supercomputer to synthesize large 4-orientable 4^2-window arrays and offer them publicly for downloading. We prototyped an algorithm for detection of the checkerboard marker fields and measured its performance. When processing input video from a cellphone camera, the algorithm visits only about 5 % of image pixels for reliable detection and the processing time is about 1 ms on a mid-range PC processor.

The Uniform Marker Field increases freedom of camera movement, especially with occluders present in the scene. The detection algorithm is efficient and real-time marker field detection will be feasible on ultramobile devices.

1 INTRODUCTION

Popular designs of fiduciary markers consist of two components: geometrical features which help to localize the marker in the processed image and features defining the identity of the marker (for example [2, 5]). That allows for placing several (or many) markers into one scene and their efficient detection [3]. Usage of several markers displaced within the scene is necessary to allow for free movement of the camera within the scene.

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Any frame of the camera must contain at least one of the markers and the marker must be large enough in order to compute precise camera location. These requirements are contradictory – the markers must be large enough to provide sufficient amount of information and at the same time they must be small enough to fit into the camera's field of view and not to be covered by any occluders present in the scene. We were looking for a way how to make the detectable markers be "present everywhere" within the picture, so that they could be detectable even in very difficult scenes.

In this work, we propose the term *Marker Field*: a planar structure whose presence can be detected in a camera image and the exact location within the field can be recognized based on any sub-area of a defined minimal size. Such sub-areas can be viewed as individual markers in the traditional sense. However, the markers are not disjoint, but they are overlapping to a large degree, in order to allow for identification from close-up views as well as from larger distances. A similar approach with slightly overlapping markers is used by the Random Dot Markers [8]. Our proposed marker field is *uniform* in the sense that the features used for detection and localization of the field and the features used for identification of individual windows (markers) within the field are uniformly distributed across the whole area of the marker field.

Our solution is based on *4-orientable n^2-window arrays* [1] – two-dimensional arrays of binary values, where each $n \times n$ sub-window is present only once, including all 4 possible rotations (Section 2 gives more information). Burns and Mitchell described the concept of orientability in De Bruijn tori; however, an algorithm for synthesis of binary De Bruijn tori was not available.

The first major contribution of our paper is that we propose a genetic algorithm for generation of such maps and we make a number of generated maps publicly available. The second major contribution is an efficient algorithm for detection of checker-board structures – such as the orientable window arrays – in a camera image. This algorithm is designed so that it can be executed on computational platforms with limited resources – such as the mobile phones and CPUs embedded in smart cameras.

2 ORIENTABLE WINDOW ARRAYS AS MARKER FIELDS

An *aperiodic (m,n)-window array* (or De Bruijn torus) [1] is an k-ary 2D array $A = (a_{i,j} \in \{0, \cdots, k-1\})$ of size $h \times w$ in which each subarray $A_{r,c}$ of size $m \times n$ occurs exactly once. If all possible subarrays are used (i.e. $(w+n-1)(h+m-1) = k^{mn}$), the (m,n)-window array is called aperiodic *perfect map* [6].

Unfortunately, when the orientation of the array is not known, the simple (m,n)-window property is not enough. It is possible that multiple rotations of the same window can occur in the ar-

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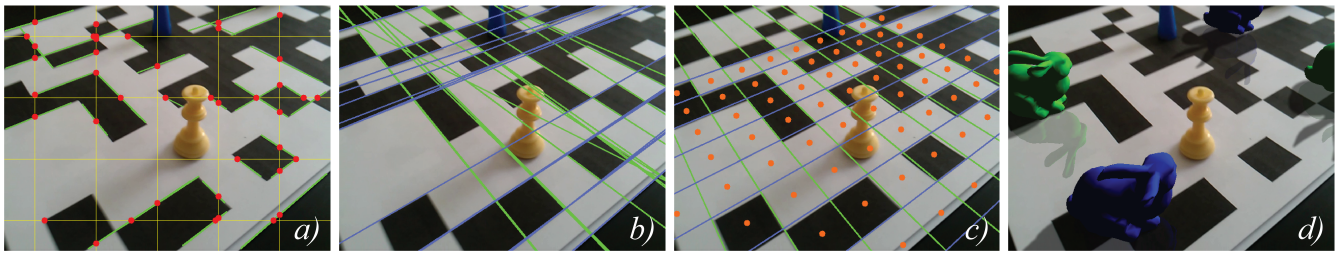


Figure 1: **Detection algorithm.** *a)* The input image is processed in scanlines (yellow). Edge points are detected (red) and edge directions found (green). *b)* The *edgels* are grouped and two vanishing points are determined (green vs. blue). *c)* Based on the vanishing points, the edgels are processed and the grid of originally perpendicular and equidistant lines is computed (green, blue) and centers of the data modules are sampled (orange). *d)* Camera location is estimated using Levenberg-Marquardt optimization minimizing the reprojection error.

ray. *Orientable window arrays* [1] solve this problem. Orientability can be gained by using certain equivalence relation for the window property [4]. We focused on 4-orientable window arrays. The 4-orientability is reasonable only for square windows; they must be unique with respect to rotation by 90°.

Contrary to the 1-orientable arrays, 4-orientable arrays are much less explored in the literature and no construction algorithms existed for binary arrays.

2.1 Synthesis of De Bruijn Tori

For binary 4-orientable aperiodic n^2-window arrays we considered arity $n = 3, n = 4, n = 5$. The theoretical upper bounds [1] of sizes for these arrays are 12×12, 127×127 and 2900×2900, respectively. Our database generated by a supercomputer contains arrays of size up to 11×11 for $n = 3$ and 92×92 for $n = 4$. Also rectangular arrays of similar $w \times h$ have been generated. The size of the field for 3^2-window arrays is very limited and the required portion of the marker field to be visible by the camera for 5^2-window arrays is unnecessarily large. For practical purposes, the 4×4 window is therefore of most interest.

Exhaustive search is not feasible as a construction method: for example, for a 4^2-window array 90×90 modules large, the area of the map to be searched for is 8100 modules and the state space is just too large (2^{8100}). The genetic algorithm is still very computationally demanding; however, it can be parallelized and executed on a large number of computation nodes. We harnessed a supercomputer of around 1 000 nodes and generated usable window arrays. One of the contributions of this work is making these arrays publicly available[1].

The genetic algorithm can be characterized by these terms:

1. For the initial population we use a number of copies of the same array, or various arrays are generated randomly.
2. The fitness function is based on the number of conflicts in the given array $f(A) = \frac{1}{c(A)+1}$.
3. The fitness threshold, where the algorithm is stopped is set to 1 (the algorithm is looking for conflict-free arrays).
4. For selecting members for the next generation, rank selection is used.
5. Mutation is defined as replacing a window with randomly generated content. The windows are selected randomly; the conflicting windows have a higher probability of being selected for replacement.

2.2 Detection of the UMF and Results

The algorithm (Fig. 1) processes the input image in widely spaced scanlines (and columns) and looks for edges by adaptive thresholding. For the found edge points, the edge orientation is determined, the points are grouped and two dominant vanishing points are computed by eigenvalue decomposition. Based on the vanishing points (and the horizon line connecting them), the accurate lines representing the edges between the marker field modules are determined [7].

According to the experiments, this algorithm visits between 5 and 6 % of image pixels for images from smartphone cameras. The algorithm is capable of processing more than 700 frames (VGA resolution) per second on a single core of a mid-range CPU (Intel® Core™i5-661) – including the rather costly homography computation. At the same time, the algorithm is very stable and invariant to rotation, perspective, blur by motion, and other factors.

[1] http://www.fit.vutbr.cz/research/groups/graph/MF

3 Conclusions

A Uniform Marker Field – unlike the conventional markers – combines the marker's geometric features used for marker localization and features used for its identification. The markers thus can overlap in the field and the field can be viewed only in part and at different scales. This allows for high freedom of camera movement while still being able to localize the camera in relation to the marker field.

The paper presents an algorithm for generation and efficient detection of suitable window arrays; in particular, 4-orientable 4^2-window arrays of practically interesting dimensions were generated by using a supercomputer and one important contribution of this paper is making them publicly available.

We plan on further optimizing and improving the detection algorithm and porting it for the most popular smarthphone platforms.

Acknowledgements

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